



A Robust EEG Brain-Computer Interface Approach for Decoding Hand and Foot Motor Imagery

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Abstract

Brain-computer interfaces (BCIs) have the capability to offer practical solutions for individuals with severe motor impairments to interact with the world. By decoding neural signals associated with motor imagery (MI), BCIs enable users to control robotic manipulators without physical movements. However, electroencephalography (EEG) signals are generally noisy and non-stationary, and therefore BCIs suffer from reliability and reproducibility in real-world applications. This work investigates the feasibility of decoding MI tasks, specifically classifying both hand and foot movements using EEG signals with high accuracy. Our proposed framework integrates advanced preprocessing, including scaling, bandpass filtering, and downsampling, along with feature extraction techniques such as Common Spatial Pattern (CSP) and Independent Component Analysis (ICA). Ten optimized machine learning (ML) classifiers were evaluated, among which Gaussian Naive Bayes (GNB), Support Vector Machine (SVM), and Decision Tree (DT) consistently exhibited superior performance. Notably, high classification accuracies (up to 100 %) for binary tasks were achieved, with minimal training data collected over four minutes per subject. Key EEG channels, particularly C3 and C4, were identified to be essential for differentiating MI tasks, adding biological relevance, and improving signal distinction. The results demonstrate the framework's robustness across various classifiers and subjects, emphasizing its potential for

real-time applications. The source code and the dataset are available at <https://github.com/MAHanafy/Motor-Imagery-WS1>

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Machine learning*; • **Hardware** → Emerging technologies.

Keywords

Brain Computer Interface(BCI), Electroencephalograph (EEG), Machine Learning, Classification, Performance evaluation

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1 Introduction

Brain-computer interface (BCI) technologies use electroencephalography (EEG) to control external devices through motor imagery (MI), where users mentally rehearse limb movements (kinesthetic or visual) without actual physical motion. These techniques are essential for individuals with severe disabilities, such as those with locked-in syndrome from amyotrophic lateral sclerosis (ALS), who cannot move their limbs. By capturing brain activity, BCIs allow users to control devices independently, which is crucial for the advancement of prosthetics for paralyzed individuals. Breakthroughs in this field facilitate the implementation of various sensorimotor functions through noninvasive EEG-based prostheses, allowing control of limbs, mobility devices, and machinery [19].

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Wet electrode EEG caps have been demonstrated to accurately and conveniently record changes in cortical activity without hindering the functionality of the BCI [13, 20]. Voltage fluctuations in the neuron's electrical gradient are captured via an EEG cap from the frontal lobe and motor cortex to gather MI data. However, raw EEG data tends to be noisy and non-stationary, posing challenges when decoding brain activity, particularly in real-time applications where energy efficiency, latency, and noise resilience are critical [5, 13]. To enhance EEG data analysis and accuracy, this research aims to pinpoint EEG signals associated with MI for more efficient preprocessing, feature extraction, and classification methods. Additionally, effective training techniques for subjects were utilized by incorporating timed intervals of rest to ensure accuracy and visual feedback.

We used a Within-Sessions (WS) approach to train our EEG data, dividing trials in each session into training and testing in an 8:2 ratio [9]. To train and enhance our BCI algorithm, public datasets for MI were taken into consideration. Plausible candidates included MILimEEG [3] and large PhysioNET datasets [17]. However, these datasets frequently have limitations that make real-time implementation difficult, like session constraints and equipment incompatibilities. To address these issues, we acquired a novel dataset that includes multiple subjects, each with multiple sessions, designed to reduce the reliance on extensive training data during each session. This paper provides benchmarking classification accuracies under WS conditions, demonstrating the potential of our approach for future implementation in real-time human-robot interaction with a mobile robot [18].

Key Contributions:

In this paper, we propose a robust system for an MI-BCI framework that extracts features that can predict MI for hand clenching (left vs. right hand) and Foot plantar flexion (left vs. right foot) via simple machine learning classifiers trained on four minutes of single-subject data. To achieve a differentiating novelty in the determination of multiclass classifications from related works, we introduce our approach of performing two separate binary classifications rather than a direct multiclass method. We also aim to eliminate confounding variables such as excess hair oil, radio waves, and the like while strategically using electrodes to accurately measure MI data for the primary motor cortex (M1). By opting for our approach, we plan to demonstrate significant accuracies without the need for large datasets and account for artifacts via 5-fold cross-validation. Our primary objective is to test various classifiers and feature extraction methods to refine MI signals and achieve an accuracy of up to 100% with low data requirements. In addition to demonstrating the feasibility of each classifier, we also analyzed the robustness of classifying each action of the hands and feet corresponding to the ICA results. Advancements in our research aim for a more intuitive approach to extracting features and preprocessing methods to integrate EEGs and BCIs.

The structure of the paper is as follows: Section 2 discusses related previous works; Section 3 describes in detail the materials and methods used; Section 4 describes the experimental setup; Section 5 presents the results and discusses their biological implications; Section 6 concludes the paper and discusses future work.

2 Related Work

In our previous works [6, 15] on BCIs, we evaluated various classifiers for MI-BCI and demonstrated that filter bank common spatial pattern (FB-CSP) outperforms common spatial pattern (CSP) in extracting discriminative motor intent features. These works established a solid foundation for integrating EEG-based binary motor execution tasks with a robotic arm to mimic a subject's arm position. We achieved an average accuracy of over 89% using SVM using extensive training data. In comparison, Choi et al. [4] focused on EEG-based motor intention robot assistive devices and achieved a peak binary classification accuracy of 86.88% using SVM. Additionally, Other related works that used large datasets and complex deep learning algorithms reported an average WS accuracy of 53.7% [9]. In contrast, our work significantly builds upon these efforts by achieving peak WS accuracies up to 100 % and average accuracy around 84% using only simple machine learning and minimal training data, highlighting the practicality and efficiency of our method.

Independent Component Analysis (ICA) has gained significant traction for its effectiveness in artifact removal and signal enhancement in MI-BCI systems. By isolating independent components from EEG data, ICA eliminates noise from sources such as ocular or muscular movements, thus preserving MI signals [20]. The integration of ICA and CSP has led to hybrid preprocessing approaches that combine artifact removal with spatial feature extraction, resulting in improved signal differentiation and classification robustness [8]. However, the comparative performance of CSP and ICA for binary MI tasks - particularly in distinguishing hand versus foot imagery, remains underexplored. Our work extends this line of inquiry and addresses gaps in the existing literature by evaluating CSP and ICA in the analysis of left versus right hands and feet.

A key innovation of our research is the use of short-duration training data. Traditional MI-BCI systems often rely on extensive training sessions and large datasets, which can be impractical for personalized applications. Recent studies have explored single-subject training datasets but continue to face challenges in achieving high classification accuracy with limited data [9]. By leveraging just four minutes of training data per subject, our framework makes MI-BCI systems more practical and accessible for real-time, personalized use. This efficiency aligns with our overarching goal of reducing training overhead without compromising system performance.

Another key contribution is the focus on classifying both hands and feet MI tasks. While most prior research has concentrated on specific limbs, few studies have addressed the complexities of distinguishing between hand and foot MI. Accurate classification of MI for both hands and feet is critical for applications in assistive devices and robotic prosthetics, where real-time control of multiple limbs is essential [4]. By analyzing classifier feasibility and robustness in distinguishing left versus right hands and feet, we contribute to the development of sophisticated, intuitive systems for real-world deployment.

Finally, achieving 100% accuracy in MI classification remains an ambitious but essential goal for MI-BCI systems. Noise, artifacts, and inter-session variability hinder real-world applications, making this objective particularly challenging. In order to address these

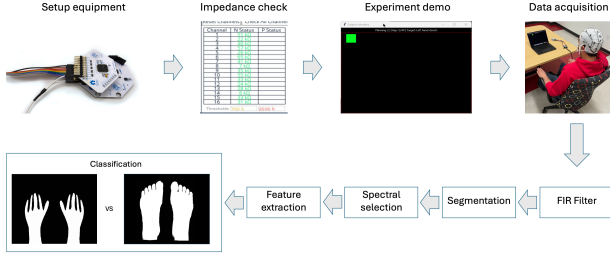


Figure 1: Proposed System

limitations, our research investigates novel feature extraction and classification techniques that enhance signal processing methods, which in turn improve system reliability and usability.

3 Materials and Methods

The proposed method aims to translate EEG signals into actionable commands that can be used to operate robots in the future. As shown in Figure 1, the entire workflow consists of preprocessing, feature extraction, and classification of EEG data. The details of these components are elaborated in the following subsections.

3.1 EEG Recording and Preprocessing

A standard OpenBCI 16-channel EEG headset [11] was used to record the EEG signals, with a sampling frequency of 125 Hz. To convert the raw signals into microvolts (μV), the data were scaled using a factor derived from the OpenBCI amplifier specifications. The mathematical definition of this scaling process is:

$$X_{\text{scaled}} = X_{\text{raw}} \times \text{SF} \quad (1)$$

where the scaling factor (SF) = $\frac{4500000}{24 \cdot (2^{23} - 1)}$.

After scaling, the signals were subjected to bandpass filtering to extract the θ (4–8 Hz), α (8–13 Hz), and β (13–30 Hz) frequency bands, which are important for MI tasks [13]. In order to minimize phase distortion, a Finite Impulse Response (FIR) filter was utilized. The filter coefficients were computed using the Hamming window method. The FIR filter can be expressed as:

$$h[k] = \left(0.54 - 0.46 \cos\left(\frac{2\pi k}{N-1}\right) \right) \cdot \left[\text{sinc}\left(\frac{2 \cdot f_{\text{high}} \cdot k}{f_s}\right) - \text{sinc}\left(\frac{2 \cdot f_{\text{low}} \cdot k}{f_s}\right) \right] \quad (2)$$

where $f_{\text{low}} = 4$ Hz and $f_{\text{high}} = 30$ and $f_s = 125$ Hz. After filtering, the signals were down-sampled in order to reduce computational complexity while preserving essential features.

The EEG data that had been preprocessed was collected into sets for each subject. The dataset for every subject i is given by $D^i = \{(X^1, y^1), \dots, (X^{N_i}, y^{N_i})\}$ which represents N_i labeled trials. For every trial j , where $1 \leq j \leq N_i$, the input matrix $X^j \in \mathbb{R}^{E \times T}$ includes the preprocessed signals of the E electrodes in T discretized time steps. The corresponding label y^j identifies the class of the trial and is drawn from a predefined set of K class labels L , representing specific imagined or executed movements. In this study, the labels of the classes

are $l_1 = \text{"Hand (Left)"}$, $l_2 = \text{"Hand (Right)"}$, $l_3 = \text{"Feet (Left)"}$, and $l_4 = \text{"Feet (Right)"}$.

In summary, the pre-processing pipeline, containing scaling, bandpass filtering, and downsampling, will ensure the extraction of clean EEG signals that are relevant. The structure of these processed signals in defined and labelled datasets will further the classification of MI tasks.

3.2 EEG Feature Extraction

To enhance the separability of EEG signals, we use a combination of band-pass filtering and Common Spatial Patterns (CSP) techniques. First, a FIR filter is used to filter the EEG signals in the α band (8–13 Hz). Because it is modulated during MI tasks and signifies event-related desynchronization (ERD) and event-related synchronization (ERS) [10, 13]. Equation of FIR filter:

$$h[n] = 2f_{\text{high}} \frac{\sin(2\pi f_{\text{high}} n)}{\pi n} - 2f_{\text{low}} \frac{\sin(2\pi f_{\text{low}} n)}{\pi n}, \quad (3)$$

where $f_{\text{low}} = 8$ Hz and $f_{\text{high}} = 13$ Hz. Following band-pass filtering, CSP is applied to extract discriminative spatial features by optimizing spatial filters that maximize the variance of one class while minimize it for another. This technique computes class-wise covariance matrices, derives spatial filters, and uses the log-transformed variances of the filtered signals as features [2, 21].

3.3 EEG Classification

The extracted features were classified using ten ML models: Linear Discriminant Analysis (LDA), Support Vector Machines (SVM), Gaussian Naive Bayes (GNB), Random Forest (RF), K-Nearest Neighbors (KNN), Decision Tree (DT), Gradient Boosting (GB), AdaBoost (AD), Logistic Regression (LR), and Multi-Layer Perceptron (MLP)[1, 14, 19]. These classifiers were designed to run in parallel, one classifier dedicated to distinguishing between hand-imagined movements (left hand vs. right hand) and another classifier dedicated to distinguishing between foot-imagined movements (left foot vs. right foot). These classifiers can be utilized for future robot commands such as right turn, left turn, acceleration, and deceleration.

- **Linear Discriminant Analysis (LDA):** finding the linear combination of features that best separates two or more classes. It ensures maximum separability by maximizing the ratio of between-class variance to within-class variance,
- **Support Vector Machines (SVM):** An SVM with a Radial Basis Function (RBF) kernel was employed to capture non-linear relationships in the data. SVMs work by identifying the hyperplane that best separates the classes in the feature space.
- **Gaussian Naive Bayes (GNB):** This probabilistic classifier applies Bayes' theorem to predict the class probabilities. It assumes that the features have a Gaussian distribution.
- **Random Forest (RF):** This ensemble method constructs multiple decision trees during training and outputs the mode of the classes for classification. It enhances classification accuracy and controls overfitting.
- **K-Nearest Neighbors (KNN):** KNN classifies a sample based on the majority class among its k -nearest neighbors in the

feature space, making it a simple yet effective non-parametric method.

- **Decision Tree(DT)**: This model splits the data into subsets based on the most significant feature at each node, creating a tree-like structure that is easy to interpret.
- **Gradient Boosting(GB)**: An ensemble technique that builds multiple weak learners (typically decision trees) sequentially. Each new improves overall performance by correcting errors made by the previous ones.
- **AdaBoost(AB)**: This ensemble technique combines multiple weak classifiers to form a strong classifier by focusing on the errors of previous classifiers and adjusting their weights accordingly.
- **Logistic Regression(LR)**: A linear model that uses the logistic function to model the probability of a binary outcome. It is extended for multi-class classification through techniques such as one-vs-rest.
- **Multi-Layer Perceptron (MLP)**: A type of artificial neural network that uses non-linear transformations to capture complex patterns in the data.

The most effective model for EEG classification was determined by comparing the results. To gain insights into the classifiers' performance in these specific tasks, Separate evaluations were conducted for the classifiers that distinguished between right-hand vs. left-hand imagined movements and left-foot vs. right-foot imagined movements.

4 Experimental Setup

This section outlines the experimental setup and procedures used to obtain EEG data from subjects. Many precautions were taken to preserve simplicity and signal quality; these precautions will be outlined in the following sections.

4.1 Subjects and Experiments Protocol

Subjects: In this experiment, EEG data were collected from 6 subjects, 4 of whom were men and 2 women. Subjects were recruited from university students and staff. The subjects were between 20 and 48 years old, with varied backgrounds, including individuals with and without prior experience in EEG-based experiments. At the beginning of the experiment, each subject signed a "Notice of Experimental Intention and Experimental Consent" to ensure their rights and interests. All the subjects were voluntary for the experiment. The subjects were identified only by their aliases "sub-01" through "sub-06". The experiments were approved by the University of Louisville Institutional Review Board (protocol number 19.0329).

Environment: The experimental environment was carefully chosen to preserve signal quality, electro-magnetic interference is a common problem in EEG signals and can significantly degrade the classification performance. This is why the subject must perform the trials far from other unnecessary devices such as computers, routers, and AC outlets. Subjects were also asked if they had any wireless headphones, smart watches, or a phone, and were asked to set them far away from the equipment.

Equipment: The equipment used in this experiment was a laptop with a 12th-generation Intel Core i7-1255U and 16 gigabytes of RAM

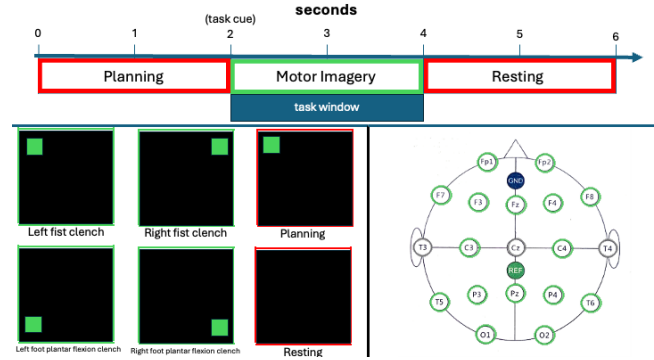


Figure 2: Collecting data

alongside a standard OpenBCI 16-channel cap [11], using a ground, a reference, and two bias electrodes. The cap was connected to an OpenBCI Cyton board with the OpenBCI Daisy module extension, allowing for a 125Hz sampling rate of 16 channels.

Preparation: Before collecting data, we must place the EEG cap on the subject's head and ensure sufficient electrode contact. This is done through the OpenBCI GUI [12]. A streaming session is started in the GUI, and a tab called "Cyton Signal", this tab measures the electrical impedance of each electrode. The GUI had a predefined range of 250 kΩ, but we aimed for <50 kΩ. The impedance starts out very high and is reduced when better contact is established between the electrode and the scalp. We do this by pouring out OpenBCI's Electrode Cap Gel into a bowl and filling a large syringe with the gel. In each electrode, the subject's scalp is lightly abraded with the syringe, moving off dirt and hair follicles and blocking contact with the scalp. The syringe is then lifted while slowly being depressed to build a bridge between the scalp and the electrode; this is done for every electrode.

4.2 Paradigm Flow

The paradigm flow, as shown in Figure 2, outlines the procedure and stimuli presented to the participant during trials. subjects are initially briefed via a demo run on the 4 different tasks: 1) **Left fist clench**, 2) **Right fist clench**, 3) **Left foot plantar flexion**, and 4) **Right foot plantar flexion**

Subjects are instructed to get into a resting state, where they will have both hands on the armrest of a chair, with the wrist protruding slightly to allow for clenching without obstruction. They are also asked to have both feet in an arched dorsiflexed state, with enough room to perform plantar flexion without obstruction. After the user has been situated, a trial is started, in which a Python program will present a black screen with a green box in one of four quadrants and a border presented in red and green. The planning stage lasts 2 seconds and is denoted by the green box in its correlated quadrant and the border color remaining red; once the border color becomes green, the task stage starts, which is another 2 seconds. Here, the user is to imagine the presented task, which is denoted by the positioning of the green box. The top left quadrant denotes left-hand clench, the top right denotes the right-hand clench, the bottom left denotes left-foot plantar flexion, and the bottom right denotes

right-foot plantar flexion. This is repeated 40 times, collecting 10 samples of each task in a single trial. The order of the tasks is randomized.

5 Results and Discussion

The performance of all classifiers was evaluated based on accuracy, F1 score, recall, and precision to determine the best model for MI classification. A 5-fold cross-validation process was employed to effectively assess the classifiers' performance. Metrics for each fold, including accuracy, F1 score, recall, and precision, were recorded and averaged. This method provides a comprehensive assessment of the model's performance, ensuring a reliable estimate of its generalization capability and mitigating the risk of overfitting.

5.1 Optimal Hyperparameters for Each Classifier

The analysis determined the best hyperparameters for all subjects and training sessions. The optimal hyperparameters for the machine learning classification methods used for the hands and feet classifiers were obtained via grid search. Table 1 summarizes the optimal hyperparameters for each classifier. Performance variability across sessions emphasizes the importance of tuning hyperparameters and selecting appropriate pre-processing techniques.

Table 1: Optimal Hyperparameters for Each Classifier

Classifier	Optimal Hyperparameters
SVM	Regularization parameter (C) of 0.1, scaled gamma, radial basis function kernel.
GNB	Smoothing parameter of $1e-09$.
RF	No maximum depth constraint, square root of the number of features for splitting, 10 estimators.
KNN	Euclidean distance metric, 3 neighbors, uniform weights.
DT	Minimum of 2 samples required to split an internal node, no maximum depth constraint, Gini criterion.
GB	Maximum depth of 3, Learning rate of 0.01, 50 estimators.
AD	Learning rate of 0.01, 50 estimators.
LR	Regularization parameter (C) of 0.1, SAGA solver.
MLP	Hidden layer size of (50, 10), maximum of 3000 iterations, ReLU activation function, L2 regularization term of 0.0001, Adam solver.
LDA	No shrinkage, the singular value decomposition solver, Tolerance of 0.0001.

5.2 Performance Metrics for Classifiers

The performance evaluation of our machine learning classification approach for MI detection of hands and feet was conducted through comprehensive grid search optimization. The results demonstrate the effectiveness of our proposed methodology across multiple performance metrics, detailed in Tables 2 and 3. Additional performance metrics, including recall and precision, are provided in Tables 4 and 5 in Appendix A.1.

The classifiers produced varying results, but most notably, SVM and GNB classifiers consistently outperformed the others. In hand MI tasks, SVM achieved an average of 84% across all subjects and sessions, with corresponding F1-scores averaging at 0.82, and peaking at 1.0. (as shown in Table 2). SVM stood out as the best-performing algorithm for hand movement classification, highlighting the effectiveness of linear decision boundaries for this task. Similarly, for

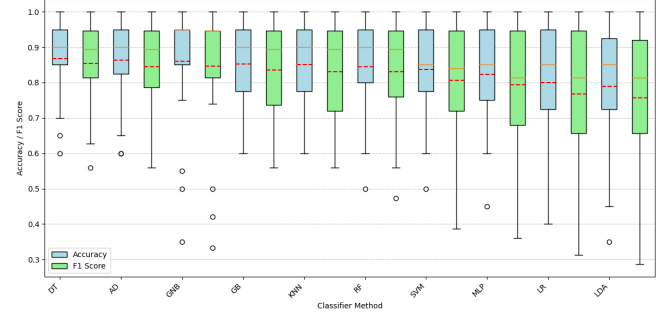


Figure 3: Classifiers comparison for hands: Accuracy and F1 scores of all subjects in all sessions

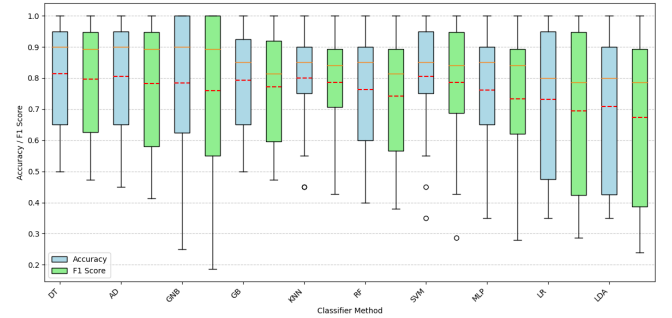


Figure 4: Classifiers comparison for feet: Accuracy and F1 scores of all subjects in all sessions

foot MI tasks, GNB achieved an average accuracy of 100% (Table 3) for certain sessions.

The most notable observations from the analysis included the robust performance of SVM, which utilized a radial basis function kernel and scaled gamma to handle non-linear decision boundaries more effectively. Its maximum precision (1.0) and recall (1.0) values set it apart from other models, showcasing its reliability at distinguishing MI tasks. The GNB classifier excelled in classifying foot MI tasks, achieving an F1-score of 1.0 in specific sessions. This reflects its ability to leverage the Gaussian distribution assumption for feature classification. Additionally, Random Forest (RF) and Gradient Boosting (GB) demonstrated reliable performance, with average accuracies exceeding 90% across both hand and foot tasks. The ability to balance model complexity and generalization was evident in their results.

Classification of feet MI tasks showed promising results, albeit slightly lower than hand MI tasks, as depicted in Figure 4. Notably, the GNB classifier revealed itself as the best choice for feet movement detection, suggesting that its probabilistic approach is well-suited for capturing distinctive patterns in feet movement data.

Figures 3 and 4 provide a visual comparison of the accuracy and F1-scores across all classifiers and sessions. The figures showcase ensemble methods (RF and GB), along with SVM and GNB, consistently outperformed other algorithms. Tables 4 and 5 in Appendix, summarize the classifiers' precision and recall metrics, while Figures 7 and 8, in Appendix, visually display the variations in precision and recall metrics across the all 18 sessions.

Table 2: Comparison of 10 different classifiers across 3 sessions (S1, S2, S3) for each of the 6 Subjects' hands using Accuracy and F1 score

Class		Subjects																	
		Subject 1			Subject 2			Subject 3			Subject 4			Subject 5			Subject 6		
		S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
DT	ACC	0.9	1.00	0.9	0.95	0.85	0.85	0.9	0.9	1.00	0.9	0.7	0.95	0.85	0.9	0.7	1.00	1.00	0.65
	F1	0.89	1.00	0.89	0.94	0.84	0.81	0.89	0.89	1.00	0.89	0.66	0.94	0.81	0.89	0.66	1.00	1.00	0.62
AD	ACC	0.9	1.00	0.9	0.95	0.8	0.85	0.95	0.9	1.00	0.9	0.65	0.95	0.85	0.9	0.7	1.00	1.00	0.6
	F1	0.89	1.00	0.89	0.94	0.76	0.81	0.94	0.89	1.00	0.89	0.57	0.94	0.81	0.89	0.66	1.00	1.00	0.56
GNB	ACC	0.95	1.00	0.95	0.95	0.8	0.9	0.95	1.00	1.00	0.95	0.75	0.9	0.95	0.95	0.35	1.00	0.95	0.5
	F1	0.94	1.00	0.94	0.94	0.76	0.86	0.94	1.00	1.00	0.94	0.74	0.89	0.94	0.94	0.33	1.00	0.94	0.42
GB	ACC	0.9	1.00	0.9	0.95	0.7	0.85	0.9	0.9	1.00	0.9	0.6	0.95	0.85	0.9	0.7	1.00	1.00	0.6
	F1	0.89	1.00	0.89	0.94	0.62	0.81	0.89	0.89	1.00	0.89	0.58	0.94	0.81	0.89	0.66	1.00	1.00	0.56
KNN	ACC	0.85	1.00	0.9	0.95	0.85	0.75	0.95	0.9	1.00	0.8	0.65	0.95	0.8	0.9	0.65	1.00	1.00	0.6
	F1	0.84	1.00	0.89	0.94	0.81	0.68	0.94	0.89	1.00	0.76	0.61	0.94	0.76	0.89	0.6	1.00	1.00	0.56
RF	ACC	0.9	1.00	0.9	0.95	0.8	0.8	0.9	0.9	1.00	0.9	0.6	0.95	0.8	0.9	0.65	1.00	1.00	0.5
	F1	0.89	1.00	0.89	0.95	0.76	0.77	0.89	0.89	1.00	0.89	0.59	0.95	0.76	0.89	0.61	1.00	1.00	0.47
SVM	ACC	0.85	1.00	0.95	0.95	0.85	0.8	0.95	0.95	0.8	0.9	0.65	0.95	0.75	0.95	0.5	1.00	0.8	0.7
	F1	0.84	1.00	0.95	0.95	0.81	0.76	0.95	0.95	0.73	0.89	0.57	0.95	0.71	0.95	0.39	1.00	0.73	0.69
MLP	ACC	0.85	1.00	0.95	0.95	0.85	0.8	0.95	0.9	0.8	0.8	0.65	0.95	0.8	0.7	0.6	1.00	1.00	0.45
	F1	0.84	1.00	0.95	0.95	0.81	0.73	0.95	0.89	0.73	0.76	0.61	0.95	0.76	0.63	0.55	1.00	1.00	0.36
LR	ACC	0.85	1.00	0.95	0.95	0.85	0.80	0.95	0.85	0.80	0.85	0.55	0.95	0.65	0.90	0.55	1.00	0.80	0.40
	F1	0.84	1.00	0.95	0.95	0.81	0.73	0.95	0.81	0.73	0.84	0.50	0.95	0.58	0.89	0.49	1.00	0.73	0.31
LDA	ACC	0.85	1.00	0.90	0.95	0.85	0.80	0.95	0.85	0.80	0.85	0.55	0.95	0.65	0.90	0.55	1.00	0.80	0.35
	F1	0.84	1.00	0.89	0.95	0.81	0.73	0.95	0.81	0.73	0.84	0.50	0.95	0.58	0.89	0.49	1.00	0.73	0.29

Table 3: Comparison of 10 different classifiers across 3 sessions (S1, S2, S3) for each of the 6 subjects' feet using Accuracy and F1 score

Class		Subjects																	
		Subject 1			Subject 2			Subject 3			Subject 4			Subject 5			Subject 6		
		S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
DT	ACC	0.95	1.00	0.90	0.95	0.65	0.90	0.95	0.70	0.90	0.60	0.55	0.90	0.65	0.95	0.85	0.65	1.00	0.90
	F1	0.95	1.00	0.89	0.95	0.63	0.89	0.95	0.68	0.89	0.56	0.53	0.89	0.55	0.95	0.84	0.63	1.00	0.89
AD	ACC	0.95	1.00	0.90	0.95	0.70	0.90	0.95	0.65	0.90	0.60	0.60	0.90	0.65	0.95	0.90	0.45	1.00	0.85
	F1	0.95	1.00	0.89	0.95	0.68	0.89	0.95	0.57	0.89	0.56	0.59	0.89	0.55	0.95	0.89	0.41	1.00	0.81
GNB	ACC	1.00	1.00	1.00	0.90	0.60	0.85	1.00	0.65	0.90	0.65	0.55	1.00	0.60	0.90	0.85	0.30	1.00	0.90
	F1	1.00	1.00	1.00	0.89	0.53	0.84	1.00	0.57	0.89	0.64	0.48	1.00	0.50	0.89	0.84	0.26	1.00	0.89
GB	ACC	0.95	1.00	0.90	0.95	0.70	0.90	0.95	0.65	0.80	0.60	0.55	0.90	0.65	0.85	0.85	0.50	1.00	0.85
	F1	0.95	1.00	0.89	0.95	0.68	0.89	0.95	0.63	0.76	0.56	0.54	0.89	0.55	0.81	0.84	0.47	1.00	0.81
KNN	ACC	0.90	1.00	0.90	0.90	0.75	0.90	0.90	0.75	0.80	0.65	0.55	0.85	0.75	0.95	0.90	0.45	1.00	0.85
	F1	0.89	1.00	0.89	0.89	0.74	0.89	0.89	0.73	0.79	0.64	0.55	0.84	0.68	0.95	0.89	0.43	1.00	0.81
RF	ACC	0.95	1.00	0.90	0.90	0.60	0.90	0.90	0.60	0.75	0.60	0.50	0.90	0.65	0.85	0.80	0.45	1.00	0.85
	F1	0.95	1.00	0.89	0.89	0.58	0.89	0.89	0.55	0.71	0.59	0.48	0.89	0.55	0.81	0.76	0.45	1.00	0.81
SVM	ACC	0.95	1.00	0.95	0.95	0.80	0.90	0.85	0.75	0.75	0.70	0.55	0.85	0.75	0.95	0.90	0.45	1.00	0.90
	F1	0.95	1.00	0.95	0.95	0.79	0.89	0.81	0.73	0.68	0.69	0.51	0.84	0.68	0.95	0.89	0.43	1.00	0.89
MLP	ACC	0.90	1.00	0.90	0.95	0.65	0.90	0.85	0.65	0.75	0.65	0.35	0.90	0.45	0.95	0.85	0.50	1.00	0.85
	F1	0.89	1.00	0.89	0.95	0.63	0.89	0.84	0.63	0.68	0.61	0.34	0.89	0.31	0.95	0.84	0.47	1.00	0.81
LR	ACC	0.95	1.00	0.90	0.95	0.50	0.95	0.65	0.45	0.75	0.60	0.45	0.80	0.45	0.95	0.90	0.45	1.00	0.85
	F1	0.95	1.00	0.89	0.95	0.42	0.95	0.57	0.35	0.68	0.58	0.43	0.79	0.31	0.95	0.89	0.39	1.00	0.81
LDA	ACC	0.90	1.00	0.90	0.90	0.35	0.95	0.65	0.35	0.75	0.60	0.40	0.80	0.40	0.95	0.90	0.45	1.00	0.85
	F1	0.89	1.00	0.89	0.89	0.32	0.95	0.57	0.24	0.68	0.58	0.38	0.79	0.28	0.95	0.89	0.39	1.00	0.81

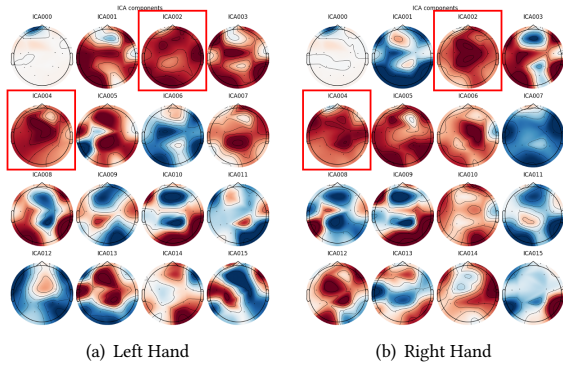


Figure 5: ICA plots of hands for one subject.

The variations observed in the performance between hand and foot movement classifications can be credited to the inherently great complexity and variability of feet movement patterns compared to those of hand movements. This observation aligns with existing literature [7, 16] on movement classification and highlights the challenges specific to lower limb movement classification.

The classifiers consistently demonstrate high precision and recall values, which indicate their ability to minimize false positives and false negatives, ensuring overall robustness. The consistent accuracy and F1 scores of further support their reliability for similar classification tasks. These classifiers, together with their associated performance metrics, offer a valuable insight for researchers working on similar datasets. The overall accuracy, F1-score, recall, and precision values emphasize the robustness and generalizability of these models, showing them as suitable choices for future applications focused on both hand and feet movement detection. Furthermore, the consistent classification performance observed across subjects suggests enhanced generalization capabilities of our proposed approach across diverse training scenarios and subject variations.

5.3 Biological Evaluation

The data was collected by EEG to measure the neuronal activity resulting from fluctuations in the ionic current within the neuron's cell membrane. Subjects were asked to imagine specific motor-related tasks to collect data on neurophysiological activity in the primary motor cortex and other areas related to the brain. These raw data are filtered through preprocessing methods to identify discriminating signals that integrate the BCI with the imagined movements of the subjects.

The strategic use of the Fz, F3, F4, C3, and C4 electrodes allowed for collecting data along the cortical region of the primary motor cortex (M1), which is responsible for signaling the initiation and execution of motor commands. Signal impedance ($<50 \text{ k}\Omega$) was also controlled by prevention of any confounding variables such as excess hair oil/products or improper fitting of the EEG cap to each subject's unique head shape to ensure adequate contact between the electrode and scalp. Obstructive signals from electromagnetic interference (EMI), such as electrical devices or strong radio frequencies, were also limited during sessions. Sessions split

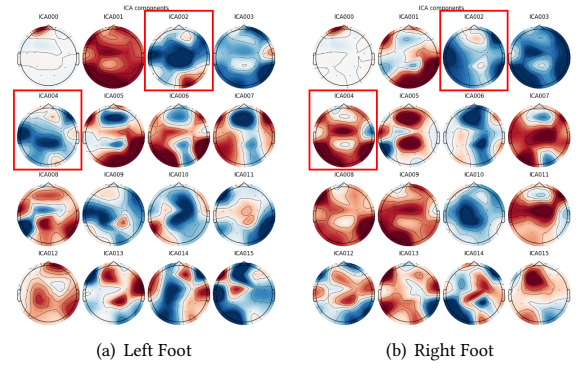


Figure 6: ICA plots of feet for one subject.

into multiple shorter sessions also produced better results, likely due to the subject's ability to regain focus and have periods of rest from performing longer periods of imagery tasks. Subjects were also informed to perform MI properly by attempting to act out the motor task in their brain and nearly preventing themselves from actually performing the motion. With this framework, we achieved impressive data accuracies up to 100% for hands vs. feet and left vs. right (hands or feet), suggesting a pattern that each classifier can deduce.

To understand which channels from the EEG corresponded to the classifiers' ability to differentiate binary MI EEG results, the data was run through an ICA to identify patterns in each channel. Observing the ICA Figures 5 and 6, we concluded that C3 (ICA002) and C4 (ICA004) channels are critical in differentiating between MI of the hands and feet since higher single amplitude is consistent throughout the hands as opposed to the feet. Research also suggests that these specific channels are viable areas for understanding the differences between each imagery task and that incorporating a full array of channels can aid in accuracy performance [16]. Additionally, channels like C3 and C4 are found to be more effective in distinguishing MI of the hands than of the feet, which can be a strong indicator of effective classification.

6 Conclusion

Our research confirms the viability of using EEG-based BCIs for real-time multi-limb control applications. We achieved high classification accuracies by leveraging CSP and a variety of classifiers, especially SVM. Our results demonstrate the robustness of our approach in decoding MI for both hand and foot movements. Importantly, our system requires only minimal training data, making it practical for real-time use. The analysis of ICA results highlights the critical role of specific EEG channels, such as C3 and C4, in differentiating between MI tasks. These findings contribute to a deeper understanding of the neural correlates of MI and inform the optimization of electrode placement for future BCI systems. Future research will extend our framework to more complex MI tasks and explore its application in real-time human-robot interaction scenarios. Furthermore, Future work will focus on expanding the dataset by including more participants to validate the generalizability of our approach.

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A Appendix

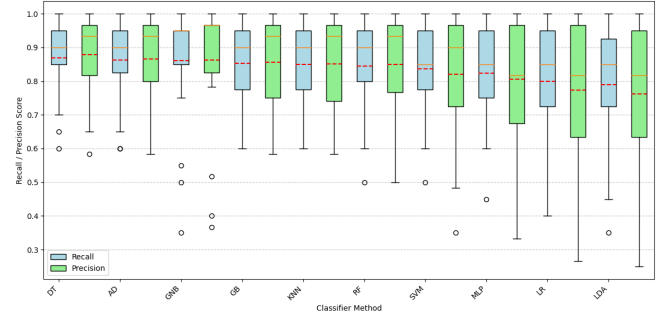


Figure 7: Classifiers comparison for hands: Recall average and precision of all subjects in all sessions

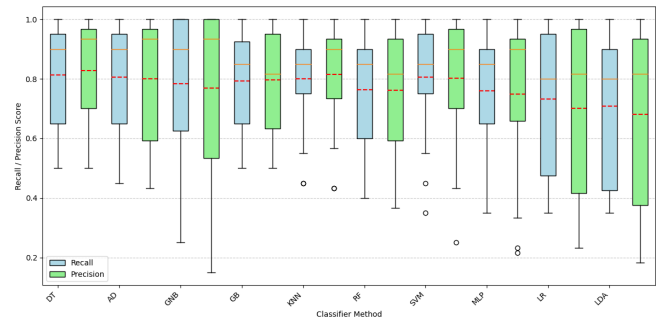


Figure 8: Classifiers comparison for feet: Recall average and precision of all subjects in all sessions

Table 4: Comparison of 10 different classifiers across 3 sessions (S1, S2, S3) for each of the 6 subjects' hands using Recall and Precision

Class		Subjects																	
		Subject 1			Subject 2			Subject 3			Subject 4			Subject 5			Subject 6		
		S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
DT	Rec	0.90	1.00	0.90	0.95	0.85	0.85	0.90	0.90	1.00	0.90	0.70	0.95	0.85	0.90	0.70	1.00	1.00	0.65
	Prec	0.93	1.00	0.93	0.97	0.90	0.82	0.93	0.93	1.00	0.93	0.65	0.97	0.82	0.93	0.68	1.00	1.00	0.70
AD	Rec	0.90	1.00	0.90	0.95	0.80	0.85	0.95	0.90	1.00	0.90	0.65	0.95	0.85	0.90	0.70	1.00	1.00	0.60
	Prec	0.93	1.00	0.93	0.97	0.78	0.82	0.97	0.93	1.00	0.93	0.60	0.97	0.82	0.93	0.68	1.00	1.00	0.58
GNB	Rec	0.95	1.00	0.95	0.95	0.80	0.90	0.95	1.00	1.00	0.95	0.75	0.90	0.95	0.95	0.35	1.00	0.95	0.50
	Prec	0.97	1.00	0.97	0.97	0.78	0.85	0.97	1.00	1.00	0.97	0.80	0.93	0.97	0.97	0.37	1.00	0.97	0.40
GB	Rec	0.90	1.00	0.90	0.95	0.70	0.85	0.90	0.90	1.00	0.90	0.60	0.95	0.85	0.90	0.70	1.00	1.00	0.60
	Prec	0.93	1.00	0.93	0.97	0.63	0.82	0.93	0.93	1.00	0.93	0.60	0.97	0.82	0.93	0.68	1.00	1.00	0.58
KNN	Rec	0.85	1.00	0.90	0.95	0.85	0.75	0.95	0.90	1.00	0.80	0.65	0.95	0.80	0.90	0.65	1.00	1.00	0.60
	Prec	0.90	1.00	0.93	0.97	0.82	0.67	0.97	0.93	1.00	0.78	0.62	0.97	0.78	0.93	0.62	1.00	1.00	0.58
RF	Rec	0.90	1.00	0.90	0.95	0.80	0.80	0.90	0.90	1.00	0.90	0.60	0.95	0.80	0.90	0.65	1.00	1.00	0.50
	Prec	0.93	1.00	0.93	0.97	0.78	0.75	0.93	0.93	1.00	0.93	0.60	0.97	0.78	0.93	0.62	1.00	1.00	0.50
SVM	Rec	0.85	1.00	0.95	0.95	0.85	0.80	0.95	0.95	0.80	0.90	0.65	0.95	0.75	0.95	0.50	1.00	0.80	0.70
	Prec	0.90	1.00	0.97	0.97	0.82	0.78	0.97	0.97	0.70	0.93	0.60	0.97	0.75	0.97	0.35	1.00	0.70	0.77
MLP	Rec	0.85	1.00	0.95	0.95	0.85	0.80	0.95	0.90	0.80	0.80	0.65	0.95	0.80	0.70	0.60	1.00	1.00	0.45
	Prec	0.90	1.00	0.97	0.97	0.82	0.70	0.97	0.93	0.70	0.78	0.65	0.97	0.78	0.63	0.55	1.00	1.00	0.33
LR	Rec	0.85	1.00	0.95	0.95	0.85	0.80	0.95	0.85	0.80	0.85	0.55	0.95	0.65	0.90	0.55	1.00	0.80	0.40
	Prec	0.90	1.00	0.97	0.97	0.82	0.70	0.97	0.82	0.70	0.90	0.52	0.97	0.57	0.93	0.48	1.00	0.70	0.27
LDA	Rec	0.85	1.00	0.90	0.95	0.85	0.80	0.95	0.85	0.80	0.85	0.55	0.95	0.65	0.90	0.55	1.00	0.80	0.35
	Prec	0.90	1.00	0.93	0.97	0.82	0.70	0.97	0.82	0.70	0.90	0.52	0.97	0.57	0.93	0.48	1.00	0.70	0.25

Table 5: Comparison of 10 different classifiers across 3 sessions (S1, S2, S3) for each of the 6 subjects' feet using Recall and Precision

Class		Subjects																	
		Subject 1			Subject 2			Subject 3			Subject 4			Subject 5			Subject 6		
		S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
DT	Rec	0.95	1.00	0.90	0.95	0.65	0.90	0.95	0.70	0.90	0.60	0.55	0.90	0.65	0.95	0.85	0.65	1.00	0.90
	Prec	0.97	1.00	0.93	0.97	0.70	0.93	0.97	0.73	0.93	0.58	0.57	0.93	0.52	0.97	0.90	0.70	1.00	0.93
AD	Rec	0.95	1.00	0.90	0.95	0.70	0.90	0.95	0.65	0.90	0.60	0.60	0.90	0.65	0.95	0.90	0.45	1.00	0.85
	Prec	0.97	1.00	0.93	0.97	0.73	0.93	0.97	0.53	0.93	0.58	0.60	0.93	0.52	0.97	0.93	0.43	1.00	0.82
GNB	Rec	1.00	1.00	1.00	0.90	0.60	0.85	1.00	0.65	0.90	0.65	0.55	1.00	0.60	0.90	0.85	0.30	1.00	0.90
	Prec	1.00	1.00	1.00	0.93	0.53	0.90	1.00	0.53	0.93	0.70	0.47	1.00	0.45	0.93	0.90	0.23	1.00	0.93
GB	Rec	0.95	1.00	0.90	0.95	0.70	0.90	0.95	0.65	0.80	0.60	0.55	0.90	0.65	0.85	0.85	0.50	1.00	0.85
	Prec	0.97	1.00	0.93	0.97	0.73	0.93	0.97	0.67	0.78	0.58	0.60	0.93	0.52	0.82	0.90	0.50	1.00	0.82
KNN	Rec	0.90	1.00	0.90	0.90	0.75	0.90	0.90	0.75	0.80	0.65	0.55	0.85	0.75	0.95	0.90	0.45	1.00	0.85
	Prec	0.93	1.00	0.93	0.93	0.80	0.93	0.93	0.77	0.83	0.70	0.57	0.90	0.67	0.97	0.93	0.43	1.00	0.82
RF	Rec	0.95	1.00	0.90	0.90	0.60	0.90	0.90	0.60	0.75	0.60	0.50	0.90	0.65	0.85	0.80	0.45	1.00	0.85
	Prec	0.97	1.00	0.93	0.93	0.63	0.93	0.93	0.55	0.72	0.63	0.53	0.93	0.52	0.82	0.78	0.47	1.00	0.82
SVM	Rec	0.95	1.00	0.95	0.95	0.80	0.90	0.85	0.75	0.75	0.70	0.55	0.85	0.75	0.95	0.90	0.45	1.00	0.90
	Prec	0.97	1.00	0.97	0.97	0.83	0.93	0.82	0.77	0.67	0.73	0.52	0.90	0.67	0.97	0.93	0.43	1.00	0.93
MLP	Rec	0.90	1.00	0.90	0.95	0.65	0.90	0.85	0.65	0.75	0.65	0.35	0.90	0.45	0.95	0.85	0.50	1.00	0.85
	Prec	0.93	1.00	0.93	0.97	0.67	0.93	0.90	0.67	0.67	0.65	0.33	0.93	0.23	0.97	0.90	0.50	1.00	0.82
LR	Rec	0.95	1.00	0.90	0.95	0.50	0.95	0.65	0.45	0.75	0.60	0.45	0.80	0.45	0.95	0.90	0.45	1.00	0.85
	Prec	0.97	1.00	0.93	0.97	0.40	0.97	0.60	0.32	0.67	0.63	0.43	0.87	0.23	0.97	0.93	0.38	1.00	0.82
LDA	Rec	0.90	1.00	0.90	0.90	0.35	0.95	0.65	0.35	0.75	0.60	0.40	0.80	0.40	0.95	0.90	0.45	1.00	0.85
	Prec	0.93	1.00	0.93	0.93	0.30	0.97	0.60	0.18	0.67	0.63	0.37	0.87	0.22	0.97	0.93	0.38	1.00	0.82